

START

READ MealQuantity, DrinkQuantity, Payment

$\text{MealSubtotal} = \text{MealQuantity} * 85$

$\text{DrinkSubtotal} = \text{DrinkQuantity} * 25$

$\text{MealDiscount} = \text{MealSubtotal} * 0.10$

$\text{DiscountedMealTotal} = \text{MealSubtotal} - \text{MealDiscount}$

$\text{PackagingFee} = \text{MealQuantity} * 5$

$\text{FinalBill} = \text{DiscountedMealTotal} + \text{DrinkSubtotal} + \text{PackagingFee}$

$\text{Change} = \text{Payment} - \text{FinalBill}$

DISPLAY(MealSubtotal)

DISPLAY(DrinkSubtotal)

DISPLAY(MealDiscount)

DISPLAY(PackagingFee)

DISPLAY(FinalBill)

DISPLAY(Change)

END

Test Case	Rice Meals	Drinks	Payments
1	3	4	₱500
2	5	3	₱600

Test Case 1

$$\text{MealSubtotal} = 3 * 85 = 255$$

$$\text{DrinkSubtotal} = 4 * 25 = 100$$

$$\text{MealDiscount} = 255 * 0.10 = 25.50$$

$$\text{PackagingFee} = 3 * 5 = 15$$

$$\text{FinalBill} = (255 - 25.50) + 100 + 15 = 344.50$$

$$\text{Change} = 500 - 344.50 = 155.50$$

Outputs:

$$\text{MealSubtotal} = 255$$

$$\text{DrinkSubtotal} = 100$$

$$\text{MealDiscount} = 25.50$$

$$\text{PackagingFee} = 15$$

$$\text{FinalBill} = 344.50$$

$$\text{Change} = 155.50$$

Test Case 2

$$\text{MealSubtotal} = 5 * 85 = 425$$

$$\text{DrinkSubtotal} = 3 * 25 = 75$$

$$\text{MealDiscount} = 425 * 0.10 = 42.5$$

$$\text{PackagingFee} = 5 * 5 = 25$$

$$\text{FinalBill} = (425 - 42.5) + 75 + 25 = 482.50$$

$$\text{Change} = 600 - 482.50 = 117.50$$

Outputs:

$$\text{MealSubtotal} = 425$$

$$\text{DrinkSubtotal} = 75$$

$$\text{MealDiscount} = 42.5$$

$$\text{PackagingFee} = 25$$

$$\text{FinalBill} = 482.50$$

$$\text{Change} = 117.50$$

Question: Why would applying the 10% discount to the entire bill produce an incorrect result?

Applying a 10% discount to the entire bill means also discounting the drinks and packaging fee's total, the problem stated that only the rice meals have a 10% discount.